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QUANTITATIVE TECHNOLOGY & FINANCIAL MARKETS SYSTEMS ARCHITECT | DERIVATIVES, FIXED INCOME, OMS AND CONTROL AUTOMATION

Technical program and solution architecture professional with 20+ years across enterprise technology and financial services. Brings direct experience designing target operating models and integration architecture for a \$25B derivatives portfolio, directing transformation of a \$54B fixed income portfolio, centralizing collateral and margining workflows, and integrating OMS controls across custodians. Hands-on AI and API architecture experience complements financial-markets systems knowledge, including deterministic validation, policy-as-code, cloud-native API orchestration and control-conscious workflow automation.

CORE EXPERTISE

Financial Markets Systems Architecture			
Derivatives Target Operating Model Design		Fixed Income Transformation	
Collateral and Margining Workflows		Order Management Systems (OMS)	
Integration Architecture	Quantitative Systems	API Architecture	
Python and Policy-as-Code		Suitability and Compliance Automation	
Architecture Audits	Rapid Prototyping	Cross-Functional Delivery	
AI Governance and Deterministic Validation			

ROLE ALIGNMENT SNAPSHOT

ABILITY TO SUPPORT TRADING-TECHNOLOGY USE CASES — DIRECT EVIDENCE

Designed derivatives integration architecture, centralized collateral and margining workflows, directed fixed income portfolio transformation, and integrated OMS controls across three custodians, providing direct experience in trading- and investment-technology operating environments.

SOFTWARE DEVELOPMENT CAPABILITY FOR QUANTITATIVE TECHNOLOGY SYSTEMS — DIRECT EVIDENCE

Architected executable Python compliance rules, cloud-native API orchestrations, a deterministic neuro-symbolic validation layer, and front-to-back product architecture,

demonstrating hands-on engineering of controlled, high-volume technology workflows.

RAPID APPLICATION DEVELOPMENT (RAD) ENGINEERING CAPABILITY — TRANSFERABLE EVIDENCE

Facilitated solution design and beta MVP release within eight weeks, led GuideBuoy AI to first beta launch in six months, and built API orchestrations handling 1,000 daily transactions.

QUANTITATIVE-FINANCE OR QUANTITATIVE-SYSTEMS DOMAIN FAMILIARITY — DIRECT EVIDENCE

Worked across derivatives and fixed income architecture, audited investment risk models and execution logic for 20+ robo-advisors, evaluated suitability algorithms, and implemented deterministic suitability logic for investment products.

PROFESSIONAL EXPERIENCE

Founder & Principal AI Architect (Agentic AI, NeSy AI, and Fiduciary Automation) | GuideBuoy AI SG Pte Ltd

August 2025 - Present - Singapore

Architecting agentic AI, neuro-symbolic AI and fiduciary automation workflows for financial-services compliance use cases.

- Architected a deterministic neuro-symbolic validation layer processing 1,000 concurrent requests with 99% accuracy to autonomously audit regulatory compliance across scam- and fraud-related financial-institution workflows.
- Built cloud-native API orchestrations handling 1,000 transactions daily, integrating LangChain and Guardrails AI to achieve 100% compliance with MAS FEAT and SRF guidelines and NIST AI RMF.
- Engineered the Consumer Guardian Engine, translating six abstract MAS Fair Dealing mandates into executable Python logic rules and reducing manual compliance review time by 100%.

Enterprise Solution Architect & Program Lead (Front Office & Regulatory Transformation) | AIA Singapore / AIA Investment Management (AIAIM)

April 2023 - April 2025 - Singapore, Singapore

Delivered front-office architecture and regulatory transformation across derivatives, fixed income, liquidity and investment-operations workflows.

- Designed the target operating model and modular integration architecture for a \$25B derivatives portfolio, enabling interoperability between BlackRock Aladdin and BNPP Calypso.
- Centralized collateral custody and outsourced critical margining workflows to BNP Paribas, replacing manual Excel-based processes with an automated system.
- Directed the conversion of a \$54B fixed income portfolio from Buy-and-Hold to Total Return, centralizing liquidity data infrastructure with BNY Mellon.

- Centralized liquidity management across 10 APAC entities and reduced SOFR payment differentials by approximately 300 bps.

Associate Director (Multi-Asset Portfolios, DPM & Digital Wealth Architecture) | Farrington Asset Management

July 2019 - December 2022 - Singapore

Worked across multi-asset portfolios, discretionary portfolio management and digital wealth architecture.

- Integrated OMS across three major custodians, automating 10+ transaction-level suitability checks daily and reducing operational errors by 90%.
- Architected deterministic suitability logic across five investment products, eliminating manual advisory bias and reducing compliance audit time by 70%.
- Grew AUM by USD \$25M in six months and secured a permanent seat on the Investment & Product Management Committee.

Lead Researcher (Algorithmic Wealth Architectures & Asset Allocation) | Synpulse Management Consulting

August 2017 - January 2018 - Singapore

Conducted research and architectural analysis across digital wealth, robo-advisory and private-banking algorithmic workflows.

- Conducted architectural audits of investment risk models and execution logic across 20+ digital wealth robo-advisors in Singapore and Hong Kong.
- Developed a proprietary ranking system using six core data points to evaluate performance and suitability algorithms across 10+ tier-1 banks in APAC.

Regional CRM & Sales Force Automation (SFA) Architect | Mondelēz International

February 2012 - September 2017 - Singapore

Architected regional CRM, sales-force automation and enterprise data-integration capabilities across APAC.

- Codified standardized retail workflows into Salesforce architecture for 1,300+ users across 16 APAC markets, supporting the \$140M Winning In Store Execution program.
- Architected a CMDM data integration layer connecting Salesforce, SAP ERP, SAP BI HANA and distributor databases.
- Managed external technology vendors to deliver implementations on schedule, realizing 61% IRR and \$102M NPV.

EDUCATION & CREDENTIALS

- Advanced Professional Certificate in Data Science and AI, Nanyang Technological University (NTU), Singapore | April 2026 - July 2026

- Master of Science (MSc) in Wealth Management, Singapore Management University (SMU) | 2016 - 2017
- Advanced Financial Analysis Series & Excel Modeling, Wall Street Training (WST) | 2016
- MBA (Master of Business Administration), North Park University | 2004 - 2005